Whitehead equivalence criterion for finitely truncated types

The Univalent Foundations Program

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Editorial note. Only the selected result is counted. Its local constructions and author proof are retained without generated proof completion.

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Allowed standard prerequisites (editorial).

- Univalent homotopy theory foundations. Univalence, function extensionality, dependent path transport, higher inductive pushouts/suspensions, truncations and basic pointed homotopy notions. The selected local code, fiber or gluing constructions are retained as author proof. Original numbered references identify these foundations.

Source reference. Original theorem 4.6.3: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1824–1824. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original section 4.6: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1680–1680. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Source reference. Original corollary 4.4.6: focused-seeds/AGP/sources/hott-book-578b85cc/extracted/book-578b85cc8d586b1677ec4335148adeb443057d24/main.labelnumbers.first-edition, source lines 1774–1774. Reference to declared foundational material or surrounding discussion; the selected local proof construction is retained in full.

Definition 0.0.1. Given a pointed map between pointed types $f : X \to Y$, we define a pointed map $\Omega f : \Omega X \to \Omega Y$ by

$$(\Omega f)(p) :\equiv {f_0}^{-1} \cdot f(p) \cdot f_0.$$

The path $(\Omega f)_0 : (\Omega f)(\mathsf{refl}_{x_0}) = \mathsf{refl}_{y_0}$, which exhibits Ωf as a pointed map, is the obvious path of type

$${f_0}^{-1} \cdot f(\mathsf{refl}_{x_0}) \cdot f_0 = \mathsf{refl}_{y_0}.$$

0.1 Whitehead's theorem and Whitehead's principle

In classical homotopy theory, a map $f : A \to B$ which induces an isomorphism $\pi_n(A, a) \cong \pi_n(B, f(a))$ for all points a in A (and also an isomorphism $\pi_0(A) \cong \pi_0(B)$) is necessarily a homotopy equivalence, as long as the spaces A and B are well-behaved (e.g. have the homotopy types of CW-complexes). This is known as *Whitehead's theorem*. In fact, the "ill-behaved" spaces for which Whitehead's theorem fails are invisible to type theory. Roughly, the well-behaved topological spaces suffice to present ∞-groupoids, and homotopy type theory deals with ∞-groupoids directly rather than actual topological spaces. Thus, one might expect that Whitehead's theorem would be true in univalent foundations.

However, this is *not* the case: Whitehead's theorem is not provable. In fact, there are known models of type theory in which it fails to be true, although for entirely different reasons than its failure for ill-behaved topological spaces. These models are "non-hypercomplete ∞-toposes" (see [1]); roughly speaking, they consist of sheaves of ∞-groupoids over ∞-dimensional base spaces.

From a foundational point of view, therefore, we may speak of *Whitehead's principle* as a "classicality axiom", akin to LEM and AC. It may consistently be assumed, but it is not part of the computationally motivated type theory, nor does it hold in all natural models. But when working from set-theoretic foundations, this principle is invisible: it cannot fail to be true in a world where ∞-groupoids are built up out of sets (using topological spaces, simplicial sets, or any other such model).

This may seem odd, but actually it should not be surprising. Homotopy type theory is the *abstract* theory of homotopy types, whereas the homotopy theory of topological spaces or simplicial sets in set theory is a *concrete* model of this theory, in the same way that the integers are a concrete model of the abstract theory of rings. It is to be expected that any concrete model will have special properties which are not intrinsic to the corresponding abstract theory, but which we might sometimes want to assume as additional axioms (e.g. the integers are a Principal Ideal Domain, but not all rings are).

It is beyond the scope of this book to describe any models of type theory, so we will not explain how Whitehead's principle might fail in some of them. However, we can prove that it holds whenever the types involved are n-truncated for some finite n, by "downward" induction on n. In addition to being of interest in its own right (for instance, it implies the essential uniqueness of Eilenberg–Mac Lane spaces), the proof of this result will hopefully provide some intuitive explanation for why we cannot hope to prove an analogous theorem without truncation hypotheses.

We begin with the following modification of original theorem 4.6.3, which will eventually supply the induction step in the proof of the truncated Whitehead's principle. It may be regarded as a type-theoretic, ∞-groupoidal version of the classical statement that a fully faithful and essentially surjective functor is an equivalence of categories.

Theorem 0.1.1. *Suppose $f : A \to B$ is a function such that*

(i) $\|f\|_0 : \|A\|_0 \to \|B\|_0$ is surjective, and

(ii) for any $x, y : A$, the function $\mathsf{ap}_f : (x =_A y) \to (f(x) =_B f(y))$ is an equivalence.

Then f is an equivalence.

Proof. Note that (ii) is precisely the statement that f is an embedding, c.f. original section 4.6. Thus, by original theorem 4.6.3, it suffices to show that f is surjective, i.e. that for any $b : B$ we have $\|\mathsf{fib}_f(b)\|_{-1}$. Suppose given b; then since $\|f\|_0$ is surjective, there merely exists an $a : A$ such that $\|f\|_0(|a|_0) = |b|_0$. And since our goal is a mere proposition, we may assume given such an a. Then we have $|f(a)|_0 = \|f\|_0(|a|_0) = |b|_0$, hence $\|f(a) = b\|_{-1}$. Again, since our goal is still a mere proposition, we may assume $f(a) = b$. Hence $\mathsf{fib}_f(b)$ is inhabited, and thus merely inhabited. □

Since homotopy groups are truncations of loop spaces, rather than path spaces, we need to modify this theorem to speak about these instead. Recall the map Ωf from Definition 0.0.1.

Corollary 0.1.2. *Suppose $f : A \to B$ is a function such that*

(i) $\|f\|_0 : \|A\|_0 \to \|B\|_0$ is a bijection, and
(ii) for any $x : A$, the function $\Omega f : \Omega(A, x) \to \Omega(B, f(x))$ is an equivalence.

Then f is an equivalence.

Proof. By Theorem 0.1.1, it suffices to show that $\mathsf{ap}_f : (x =_A y) \to (f(x) =_B f(y))$ is an equivalence for any $x, y : A$. And by original corollary 4.4.6, we may assume $f(x) =_B f(y)$. In particular, $|f(x)|_0 = |f(y)|_0$, so since $\|f\|_0$ is an equivalence, we have $|x|_0 = |y|_0$, hence $|x = y|_{-1}$. Since we are trying to prove a mere proposition (ap_f being an equivalence), we may assume given $p : x = y$. But now the following square commutes up to homotopy:

$$\begin{array}{ccc} \Omega(A, x) & \xrightarrow{- \cdot p} & (x =_A y) \\ {\scriptstyle \Omega f}\downarrow & & \downarrow{\scriptstyle \mathsf{ap}_f} \\ \Omega(B, f(x)) & \xrightarrow[- \cdot f(p)]{} & (f(x) =_B f(y)). \end{array}$$

The top and bottom maps are equivalences, and the left-hand map is so by assumption. Hence, by the 2-out-of-3 property, so is the right-hand map. □

Theorem 0.1.3. *Suppose A and B are n-types and $f : A \to B$ is such that*

(i) $\|f\|_0 : \|A\|_0 \to \|B\|_0$ is a bijection, and
(ii) $\pi_k(f) : \pi_k(A, x) \to \pi_k(B, f(x))$ is a bijection for all $k \geq 1$ and all $x : A$.

Then f is an equivalence.

Condition (i) is almost the case of (ii) when $k = 0$, except that it makes no reference to any basepoint $x : A$.

Proof. We proceed by induction on n. When $n = -2$, the statement is trivial. Thus, suppose it to be true for all functions between n-types, and let A and B be $(n+1)$-types and $f : A \to B$ as above. The first condition in Corollary 0.1.2 holds by assumption, so it will suffice to show that for any $x : A$, the function $\Omega f : \Omega(A, x) \to \Omega(B, f(x))$ is an equivalence.

Since $\Omega(A, x)$ and $\Omega(B, f(x))$ are n-types we can apply the induction hypothesis. We need to check that $\|\Omega f\|_0$ is a bijection, and that for all $k \geq 1$ and $p : x = x$ the map $\pi_k(\Omega f) : \pi_k(x = x, p) \to \pi_k(f(x) = f(x), \Omega f(p))$ is a bijection. The first statement holds by assumption, since $\|\Omega f\|_0 \equiv \pi_1(f)$. To prove the second statement, we generalize it first: we show that for all $y : A$ and $q : x = y$ we have $\pi_k(\mathsf{ap}_f) : \pi_k(x = y, q) \to \pi_k(f(x) = f(y), \mathsf{ap}_f(q))$. This implies the desired statement, since when $y :\equiv x$, we have $\pi_k(\Omega f) = \pi_k(\mathsf{ap}_f)$ modulo identifying their base points $\Omega f(p) = \mathsf{ap}_f(p)$. To prove the generalization, it suffices by path induction to prove it when q is refl_a. In this case, we have $\pi_k(\mathsf{ap}_f) = \pi_k(\Omega f) = \pi_{k+1}(f)$, and $\pi_{k+1}(f)$ is an bijection by the original assumptions. □

References

[1] Jacob Lurie. *Higher topos theory.* Number 170 in Annals of Mathematics Studies. Princeton University Press, 2009. arXiv:math.CT/0608040.